

Greedy Packing of Nested Rings: Superadditivity, Placement Rules, and a Tribonacci Threshold

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Abstract

We study packings of annuli (“rings”) of common width into a disk, where a ring may nest inside the hole of a strictly larger one – a circular, selection-oriented relative of the Recursive Circle Packing Problem. The two natural objectives, number of placed rings and total contact area, genuinely diverge: area is a superadditive function of the radius, cardinality is not. For *superincreasing* radii (each exceeding the sum of all smaller ones) we prove that the descending greedy maximizes every positive, strictly increasing, superadditive objective, and – our main structural theorem – that the placement rule is irrelevant: any choice among feasible containers yields the lexicographically maximal feasible set, for containers of arbitrary shape and in every dimension. Both hypotheses are sharp: placement irrelevance holds unconditionally for at most three rings and fails at four, and *twin instances* rule out every placement rule that is a function of the observable state. Measuring the violation of superincreasingness by $\rho = \max_i(\sum_{j>i} r_j)/r_i$, the additive relaxation of the model has universal threshold exactly $\rho = 1$, while the geometric model resists further: we prove, with no tangency idealization, that the four-ring rigid family of counterexamples has infimum exactly the Tribonacci constant $T \approx 1.83929$, and we conjecture that the geometric threshold is exactly T . We develop a supporting program towards the conjecture – exact positive-width curves with global minimum $13/7$, closed profile thresholds, and blocking walls for generic containers with metallic-mean floors – and complement it with a phase diagram for the divergence of the objectives and a decoupling of hardness into a geometric layer and a combinatorial (subset-sum) layer, of which superincreasingness eliminates exactly the latter.

1 Introduction

Drop a batch of squid rings into a frying pan and a natural optimization problem appears: place as many rings as possible flat on the pan – or, if one cares about searing, maximize the total ring–pan contact area – exploiting the fact that a sufficiently small ring fits inside the hole of a larger one. Abstracted, this is the problem of packing annuli of common width into a disk with recursive nesting, a circular-container relative of the *Recursive Circle Packing Problem* (RCPP) introduced by Pedrosa, Cunha and Tavares [1] to model the “telescoping” of tubes in shipping containers, and attacked exactly by Gleixner, Maher, Müller and Pedrosa [2]. That literature is algorithmic: it packs a fixed set of rings into as few rectangular containers as possible. Here we

*Independent researcher. javier.aguilar@sapira.ai. Computational verification scripts for every numerical claim, together with extended proofs and the verification reports cited in the appendix, are available in the accompanying repository, <https://github.com/JaviMaligno/calamares> (python code/run.all.py reproduces every verification; the arXiv version will pin the exact commit).

ask structural questions instead: which objectives does the natural greedy algorithm optimize, when is the choice of *where* to put each ring irrelevant, and what conditions on the radii govern the answers.

Our starting point is a classical phenomenon in one dimension. For bin packing with *divisible* item sizes, Coffman, Garey and Johnson [3] proved that First Fit Decreasing is optimal. We are not aware of an analogue for geometric packing of disks, let alone with nesting. We prove one, with *superincreasing* radii (each radius exceeding the sum of all smaller ones) playing the role of divisibility, and we determine exactly how far the hypothesis can be relaxed – with a surprising answer: the additive relaxation of the model has threshold exactly 1, while the geometric model resists up to (conjecturally, exactly) the Tribonacci constant.

Contributions: main theorems. (1) A *superadditivity dichotomy*: under superincreasing radii the selection greedy is optimal for every positive, strictly increasing, superadditive value function of the radius – contact area is one – while cardinality (not superadditive) admits counterexamples even under superincreasing radii (Section 3). The selection statement is best seen as a general fact about downward-closed set systems with superincreasing weights, in the lineage of greedy analyses on independence systems [13, 14] and of the superincreasing knapsack [15]; the geometric content of this paper begins with the next item. (2) A *placement-obliviousness theorem*: under superincreasing radii, *any* rule for choosing among feasible containers yields the lexicographically maximal feasible set; the proof is container-shape- and dimension-agnostic (Section 4). (3) *Sharpness*: placement obliviousness holds unconditionally for $n \leq 3$ rings and fails at $n = 4$; twin instances with an identical prefix rule out every state-based placement rule, deterministic or randomized (Section 5). (4) *Thresholds*: the additive model has universal threshold exactly $\rho = 1$; the geometric four-ring rigid subfamily has infimum exactly the Tribonacci constant T , proved with no idealization (Theorem 17, full proof in Appendix B), and we conjecture the geometric threshold equals T (Section 6).

Contributions: the program towards the Tribonacci threshold. (5) *The width program*: the blocked-exchange infimum with relative width $\omega > 0$ is governed by the deformed Tribonacci cubic $\alpha^3 = (1 + \omega)(\alpha^2 + \alpha + 1)$; the full lower-bound curve is in closed form and its global minimum is exactly $13/7$, attained at a rational corner (Section 7); the combinatorial profile threshold is closed for every profile size, with golden plateau and exact Tribonacci crossing $\omega_T = 1/T - 1/2$. (6) *Generic containers*: an exact corona criterion for wall-tangent packings (with two corrective counterexamples and a $k = 5$ “pentagram” certificate), and a hierarchy of blocking walls – occupant holes, “blocking costs the slack”, and the double pocket – yielding metallic-mean floors (silver ratio $1 + \sqrt{2}$, golden line $\varphi^2 - (\varphi/2)\omega$) that push every blocked exchange with extra occupants above T – for all widths $\omega < 0.9626$ with one extra occupant, up to 0.624 and 0.896 with two and three, and for all widths with four or more (Section 8). The remaining gaps of the program are stated explicitly there and in Open Problem 22. (7) A phase diagram for the divergence of the two objectives, and a decoupling of hardness into a geometric and a combinatorial layer, the latter with an explicit PARTITION reduction (Sections 9–10).

2 Model and preliminaries

Definition 1 (Rings and placements). Fix a width $w > 0$. A *ring* of outer radius $r > 0$ is an annulus with inner (hole) radius $\max(0, r - w)$; if $r \leq w$ it is a solid disk with no hole. Let

$K \subset \mathbb{R}^d$ be a compact *container* (the *pan*; our motivating case is a disk of radius R in the plane, $d = 2$, but Theorems 6 and 9 hold for arbitrary K and d , with rings read as spherical shells). A *placement* of a subset S of the rings is a forest: each placed ring has as parent either the pan or a placed ring with $r_{\text{child}} \leq r_{\text{parent}} - w$ (it sits inside the parent's hole); the children of a common parent must be packable as pairwise-disjoint balls (by outer radius) inside the parent's region — K itself, or the hole ball of radius $r - w$. Feasibility is a property of the assignment (siblings may be rearranged freely inside their container).

Definition 2 (Objectives; superincreasing radii). For radii $r_1 > r_2 > \dots > r_n$, the *contact area* of a placed ring is $a(r) = \pi(r^2 - \max(0, r - w)^2)$; the two objectives are $N = |S|$ and $A = \sum_{i \in S} a(r_i)$ over feasible S . The radii are *superincreasing* if $r_i > \sum_{j > i} r_j$ for all i ; the violation is measured by $\rho = \max_i (\sum_{j > i} r_j) / r_i$. The *lex-max* set L is defined greedily: scanning $i = 1, \dots, n$, include i whenever $(L \cap \{1, \dots, i - 1\}) \cup \{i\}$ is feasible. Since feasibility is downward closed, L is the unique lexicographically maximal feasible set, and the *selection greedy* (add ring i if the resulting set remains feasible) computes it. Note that the selection greedy queries a *set-feasibility oracle*; it is an information bound, not an efficient algorithm — deciding sibling packability is itself hard (Section 10). The placement greedies below, by contrast, only ever attempt concrete insertions. A *descending greedy with placement rule* processes rings in decreasing radius and places each into *some* feasible container (the pan or the hole of an already placed ring), skipping it only if no container admits it.

Lemma 3 (Row lemma). *Balls with radii $\rho_1, \dots, \rho_k$ satisfying $\sum_j \rho_j \leq C$ pack inside a ball of radius C : place them tangent in a row along a diameter. Consequently, if a ball of radius m is removed from any packing, any collection of balls of total radius at most m can be inserted into the vacated ball without disturbing the rest.*

Proof. With centers on a diameter at coordinates $x_j = -C + 2 \sum_{l < j} \rho_l + \rho_j$, each satisfies $|x_j| \leq C - \rho_j$. $\square$

3 The superadditivity dichotomy

Lemma 4. *The contact area a is increasing and superadditive on $(0, \infty)$: $a(x) + a(y) \leq a(x + y)$.*

Proof. Monotonicity is clear. For $x, y \geq w$: $\pi w(2x - w) + \pi w(2y - w) = \pi w(2(x + y) - 2w) \leq \pi w(2(x + y) - w)$. For $x, y \leq w$ with $x + y \leq w$: $x^2 + y^2 \leq (x + y)^2$. For $x, y \leq w < x + y$: $x^2 + y^2 \leq w(x + y) \leq 2w(x + y) - w^2$, using $x^2 \leq wx$, $y^2 \leq wy$ and $x + y \geq w$. For $x \leq w \leq y$: $\pi x^2 + \pi w(2y - w) \leq \pi w(2x + 2y - w)$ since $x^2 \leq 2wx$. $\square$

Lemma 5 (Lexicographic dominance). *Let the radii be superincreasing and let v be any positive, increasing, superadditive function of the radius. If S, T are feasible sets and the first index (in decreasing radius order) at which they differ belongs to S , then $\sum_S v(r_i) \geq \sum_T v(r_i)$, with equality only if $T \subseteq S$.*

Proof. Let i be the first differing index and $P = S \cap \{1, \dots, i - 1\} = T \cap \{1, \dots, i - 1\}$. If $T \setminus P = \emptyset$ then $T \subseteq S$ and monotonicity of the sum under inclusion suffices. Otherwise $T \setminus P \subseteq \{i + 1, \dots, n\}$, so $\sum_{T \setminus P} r_j < r_i$ by superincreasingness, and by iterated superadditivity and monotonicity $\sum_{T \setminus P} v(r_j) \leq v(\sum_{T \setminus P} r_j) < v(r_i)$. Hence $\sum_T v = \sum_P v + \sum_{T \setminus P} v < \sum_P v + v(r_i) \leq \sum_S v$. $\square$

Theorem 6 (Selection greedy; arbitrary container, any dimension). *With superincreasing radii, the selection greedy computes the lex-max feasible set and therefore maximizes $\sum_{i \in S} v(r_i)$ for every positive, strictly increasing, superadditive v ; in particular it maximizes the contact area A .*

Proof. Feasibility is downward closed, so the selection greedy computes L (if it could add $i \notin L$, any maximal feasible extension would lexicographically beat L). Lemma 5 shows L dominates every feasible set. $\square$

Remark 7 (Positioning). Theorem 6 is at heart a statement about arbitrary downward-closed set systems with superincreasing weights, and as such it belongs to the classical circle of ideas around greedy algorithms on independence systems [13, 14] and the superincreasing knapsack, whose lexicographic structure was analysed by Gupte [15]. The hypotheses on v are all needed: positivity rules out $v \equiv -1$ (superadditive and non-decreasing, yet maximized by the empty set), and strict monotonicity is what makes the dominance in Lemma 5 strict. The geometric novelty of this paper lies in Theorem 9, where the exchange argument must respect the geometry of nested containers.

Proposition 8 (Cardinality is not rescued). *There are superincreasing instances on which every descending greedy is suboptimal for N . Example (verified): pan $R = 10$, $w = 4.8$, radii $\{9.95, 5.0, 4.3, 0.6\}$. Greedy places $\{9.95, 5.0\}$ ($N = 2$, optimal area), while $\{5.0, 4.3, 0.6\}$ packs in a row in the pan ($N = 3$). The obstruction is structural: N corresponds to $v \equiv 1$, which is not superadditive, and Lemma 5 fails for it.*

4 Placement obliviousness under superincreasing radii

Theorem 9 (Placement rule irrelevance). *Let the radii be superincreasing (weakly: $r_i \geq \sum_{j>i} r_j$ suffices) and let $K \subset \mathbb{R}^d$ be an arbitrary container. Every descending greedy, with an arbitrary rule for choosing among feasible containers, places exactly the lex-max feasible set. In particular best fit, worst fit and random placement are all optimal for every positive, strictly increasing, superadditive objective.*

Proof. Let F denote the greedy's placement just before processing ring i , with placed set G , and suppose $G \cup \{i\}$ is feasible via a witness placement P ; it suffices to show some container of F admits i (the converse direction is immediate, and then greedy's set equals L by induction as in Theorem 6). The available containers – the pan plus one hole per ring of G – coincide in F and P , and in P the smallest ring i parents nobody.

Let m be the largest ring of G assigned different containers by F and P (if none, P restricted to G equals F and i 's container in P witnesses admission). Let u be F 's container for m and v be P 's. Since the greedy processes rings in decreasing order, when F placed m the occupants of u were exactly the rings larger than m that F assigns to u ; by maximality of m these coincide with the rings larger than m that P assigns to u , and F certified that this set together with m packs in u . Build P' from P : (i) every ring smaller than m that P kept in u is moved into the ball of radius r_m vacated by m inside v – they fit in a row by Lemma 3, because their total radius is at most $\sum_{j:r_j < r_m} r_j \leq r_m$, and they touch nothing else in v ; (ii) m moves to u , joining exactly the larger-than- m rings whose compatibility F certified. Nesting constraints persist: u and v are the pan or holes of rings larger than m , which do not move, and every ring moved into v is smaller than r_m , which fitted in v . Thus P' is a feasible witness agreeing with F on all rings of radius $\geq r_m$: the first disagreement strictly shrinks. Iterating at most $|G|$ times yields a

witness P^* agreeing with F on all of G and placing i in some container c^* ; since the occupants of c^* in P^* are exactly those in F , the greedy admits i there. $\square$

Remark 10. The proof uses only (a) downward closure of packings, (b) the row lemma applied *inside the vacated ball*, and (c) the total-radius bound from superincreasingness. Nothing refers to the shape of K or the dimension: the theorem covers rectangular pans (griddles) and spherical shells nested in arbitrary containers in $\mathbb{R}^d$ verbatim. Computational corroboration of the strongest consequence: on 100 random superincreasing instances, best-fit, worst-fit and random placement produced identical, optimal outcomes without exception.

5 Sharpness: the $n = 4$ transition and twin instances

Proposition 11 ($n \leq 3$, disk pan). *For a disk pan and arbitrary radii, every descending greedy on at most three rings places the lex-max set.*

Proof. The only branching point is ring r_2 , when both the pan ($r_1 + r_2 \leq R$; two circles fit in a disk iff their radii sum to at most its radius) and the hole of r_1 ($r_2 \leq r_1 - w$) admit it. Let G be the greedy's set after r_2 and suppose $G \cup \{r_3\}$ is feasible via a witness P ; we check that the greedy's placement F admits r_3 in each of the four combinations of choices. If F nested r_2 : should P nest it too, the containers of F and P have identical occupants and P 's home for r_3 works in F ; should P keep r_2 in the pan, then $r_1 + r_2 \leq R$, so $r_1 + r_3 < R$ and F 's pan (holding only r_1) admits r_3 . If F kept r_2 in the pan: should P do likewise, the placements agree; should P nest r_2 , then $r_2 \leq r_1 - w$, hence $r_3 < r_2 \leq r_1 - w$ and F 's *empty* hole of r_1 admits r_3 . In every case r_3 is admitted, and rings outside the lex-max set are never admissible, so the greedy's set is the lex-max set. $\square$

Lemma 12 (Cap confinement). *Circles of radii 10, 4.9, 4.8 do not pack in a disk of radius 15.*

Proof. Write c_A, c_X, c_Y for the centers, measured from the pan center. Then $|c_A| \leq 5$, $|c_X| \leq 10.1$, $|c_Y| \leq 10.2$, $|c_A - c_X| \geq 14.9$, $|c_A - c_Y| \geq 14.8$, $|c_X - c_Y| \geq 9.7$. From $|c_A| + |c_X| \geq 14.9$, $a := |c_A| \in [4.8, 5]$. Let $u = -c_A/|c_A|$. Then $c_X \cdot u = (|c_X - c_A|^2 - |c_X|^2 - a^2)/(2a) \geq (120 - a^2)/(2a) \geq 9.5$ (decreasing in a), so the transverse part of c_X has norm at most $\sqrt{10.1^2 - 9.5^2} \leq 3.43$; similarly $c_Y \cdot u \geq (115 - a^2)/(2a) \geq 9.0$ with transverse norm at most $\sqrt{10.2^2 - 9^2} = 4.8$. Hence $|c_X - c_Y|^2 \leq 10.1^2 + 10.2^2 - 2(9.5 \cdot 9.0 - 3.43 \cdot 4.8) \leq 67.98$, i.e. $|c_X - c_Y| \leq 8.25 < 9.7$, a contradiction. $\square$

Theorem 13 (Failure at $n = 4$). *Placement obliviousness fails for four rings: with $R = 15$, $w = 0.3$ and radii $\{10, 5, 4.9, 4.8\}$ (all four form the lex-max set, witnessed by the pan pair $\{10, 5\}$ at exact tangency and the hole pair $\{4.9, 4.8\}$ filling the hole 9.7 exactly), best fit nests the 5, forces the 4.9 into the pan, and blocks the 4.8 everywhere by Lemma 12 and the exact two-circle conditions; worst fit places all four.*

Theorem 14 (Twin instances; no state-based rule). *Let $R = 15$, $r_1 = 10$, $r_2 = 5$, $w = 0.505$ (hole 9.495), and*

$$I_1 = \{10, 5, 4.99, 4.50\}, \quad I_2 = \{10, 5, 4.76, 4.74\}.$$

In I_1 the correct placement of the 5 is the pan (best fit fails, worst fit succeeds); in I_2 it is the hole (worst fit fails, best fit succeeds). At the decisive step the observable state – containers, capacities,

occupants, incoming ring, R and w – is identical in I_1 and I_2 . Consequently no deterministic placement rule that is a function of the state attains the lex-max set on all instances, and for every randomized rule some instance is failed with probability at least $1/2$.

Proof ingredients. Feasibilities are two-circle-exact except for two triples, both settled rigorously: $\{10, 4.99, 4.50\}$ does not pack in the disk of radius 15 (cap confinement as in Lemma 12 yields $|c_X - c_Y| \leq 8.80 < 9.49$), and the pan pair $\{10, 5\}$ is diametrically rigid, whence no third circle of radius ≥ 4.7 fits (confinement again: for $z = 4.76$, $|c_z - c_B| \leq \sqrt{204.86 - 20 \cdot 8.8} \leq 5.37 < 9.76$). Packability of $\{10, 4.76, 4.74\}$ in radius 15 is exhibited by an explicit boundary configuration (angles 180° , 30.8° , -32.0° ; all pairwise constraints verified). The key phenomenon enabling the twins is that triple-packability is *not monotone in the sum*: the unbalanced pair $\{4.99, 4.50\}$ (sum 9.49) fails beside the 10 while the balanced pair $\{4.76, 4.74\}$ (sum 9.50) succeeds. $\square$

Remark 15. The obstruction is informational, not existential: following any witness of the lex-max set is itself a legal greedy execution (a ring outside the lex-max set is never admissible), so some execution always succeeds; what cannot exist is a state-based rule that finds it. For rules with access to the full input the question becomes one of computational complexity and remains open (Section 12).

6 Thresholds: additive sharpness and the Tribonacci floor

Theorem 16 (Additive model: threshold exactly 1). *Replace sibling feasibility by its additive surrogate (a set of siblings fits iff their radii sum to at most the container capacity; exact for at most two circles and sufficient in general by Lemma 3). Then the universal threshold for placement obliviousness is exactly 1: obliviousness holds for all instances with $\rho \leq 1$ (the proof of Theorem 9 applies verbatim), while for every $\varepsilon > 0$ some instance with $\rho \leq 1 + \varepsilon$ fails – individual instances with larger ρ may of course still succeed. Indeed, with r_1 fixed, $s = R - r_1 \in (\frac{2}{3}(r_1 - w), r_1 - w]$, $w > s/2$, take $r_2 = s$ and a pair $r_3 \approx r_4 \approx s/2$ with $r_3 + r_4 = s + \varepsilon$: the witness fills the pan with $\{r_1, r_2\}$ and the hole with the pair, while best fit nests r_2 , sends r_3 to the pan, and blocks r_4 additively ($r_1 + r_3 + r_4 > R$; $r_2 + r_4 > r_1 - w$; $r_4 > r_2 - w$). All ratios tend to 1; verified instances reach $\rho = 1.03$ and 1.02 .*

The four-ring counterexample family carries a genuine floor at the Tribonacci constant. An earlier version of this statement relied on a tangency idealization ($r_3 \rightarrow r_2$, $w \rightarrow 0$); the following theorem removes it entirely: the rigid configuration is not assumed but *emerges* as the extremal case of a concavity argument.

Theorem 17 (Rigid-subfamily Tribonacci floor, no idealization). *Normalize $r_1 = 1$ and write $t = r_2$, $u = r_3$, $v = r_4$, $\omega = w > 0$. Let $\mathcal{F}$ be the family of four-ring instances with (F1) diametral pan $R = 1 + t$, (F2) $u + v \leq 1 - \omega$ (the witness pair fits the hole of r_1), and (F3) the triple $\{1, u, v\}$ does not pack in the disk of radius $1 + t$. Then every instance of $\mathcal{F}$ satisfies $\rho > T$, strictly, for every width $\omega > 0$ and all strict radii $v < u < t < 1$; moreover $\inf_{\mathcal{F}} \rho = T$ and the infimum is not attained. The family is nonempty only for $t < t^* = 1/T$, and it contains the $n = 4$ counterexample and the twin I_1 of Theorem 14.*

Proof sketch (complete proof in Appendix B). For two circles of radii a, b internally tangent to the wall of a disk of radius R , the minimal angular separation $\theta(a, b)$ obeys the half-angle identity $\sin^2(\theta/2) = f(a)f(b)$ with $f(x) = x/(R - x)$. Placing the triple $\{1, u, v\}$ wall-tangent with the

unit ring in the middle packs it whenever $\theta(1, u) + \theta(1, v) + \theta(u, v) \leq 2\pi$ (the two non-adjacent arguments are separated by monotonicity of θ), and this angular condition reduces *algebraically* to

$$\psi(u) + \psi(v) \geq \tau, \quad \psi(x) = \sqrt{\frac{t-x}{x}}, \quad \tau = \frac{t}{\sqrt{1+t}},$$

(an algebraic *sufficient* condition for packing, which is all the contrapositive needs: (F3) forces $\psi(u) + \psi(v) < \tau$). In the coordinates $\alpha = \psi(u)$, $\beta = \psi(v)$ the sum $u + v = U(\alpha) + U(\beta)$ with $U(z) = t/(1 + z^2)$, and U is concave on $[0, \tau]$ exactly when $\tau \leq 1/\sqrt{3}$, i.e. $t \leq (1 + \sqrt{13})/6 = 0.7676\dots$, which covers the relevant range with room to spare; hence the infimum of $u + v$ over the blocked (open) region sits at the *corner* of its closure, $u = t$, $v = b(t)$, where $b(t) = t(1 + t)/(1 + t + t^2)$ is the Descartes pocket of the rigid pair. This gives the two pressures $t + b(t) < u + v \leq 1 - \omega < 1$ (the first one strict, whence the strict floor and the non-attainment), whose compatibility is precisely the cubic $t^3 + t^2 + t < 1$, and then $\rho \geq (u+v)/t > 1 + (1+t)/(1+t+t^2) =: L(t)$ with L strictly decreasing and $L(t^*) = T$. Exactness of the pocket at the rigid pair (needed only for the direction $\inf \leq T$) is a rigidity computation: the constraint left-hand side factors exactly as $4(t^2 + t + 1)(b(t) - v)$; and the approximating family lives genuinely inside $\mathcal{F}$ by a closure-and-monotonicity lemma for the packing region (compactness of witnesses), which shows the blocked interval in u is half-open with an attained endpoint. $\square$

Remark 18 (Slack pans). The hypothesis $R = r_1 + r_2$ can be relaxed to $R \geq r_1 + r_2$ at no cost: non-packability is antitone in the container (a packing in a smaller disk is a packing in a larger one after translation), so (F3) at radius R implies (F3) at radius $r_1 + r_2$ and Theorem 17 applies verbatim; the infimum remains T . Slack in the pan never helps the blocker — what it changes is the set of possible *occupants*, the subject of Section 8.

Conjecture 19 (Tribonacci threshold). *In the geometric model, placement obliviousness holds for all instances with $\rho < T$ and fails for suitable instances at every $\rho > T$; that is, the geometric threshold is exactly the Tribonacci constant, and $T - 1$ measures the extra protection that disk geometry (Descartes pockets) grants the greedy over additive bookkeeping. Evidence beyond Theorem 17 and Sections 7–8: the additive $\rho \rightarrow 1$ family is rescued geometrically (its small rings, at half the instance scale, sit well below the pocket $\approx 0.43 \times \text{scale}$); structured searches restricted to $\rho < 1.8$ (120 five-ring instances, best and worst fit versus the lex-max oracle) and more than 900 further runs produced no failure below T .*

7 The width program: profile thresholds and the 13/7 corner

Towards Conjecture 19, the proof of Theorem 9 localizes all difficulty in a single *exchange step*: m is the largest ring that the greedy F and a witness P place in different containers $u = c_F(m)$, $v = c_P(m)$, and the set S of rings smaller than m that P keeps in u must be reinserted using the resources freed by moving m to u : the vacated disk D_m (radius r_m), the hole H_m (capacity $r_m - w$, travelling with m), nesting, and the geometry of v . Normalize $r_m = 1$ and $\omega = w/r_m$. Everything ρ imposes on S is the pair of inequalities $\sum S \leq \rho$ and $\sum_{l > j} x_l \leq \rho x_j$; a lower bound for ρ over all *blocked* exchanges is therefore a lower bound for any failure of placement obliviousness.

The combinatorial layer is closed. For two-ring profiles the blocked threshold is exactly $\rho_2^*(\omega) = \max(1, 2(1 - \omega))$; in particular no two-ring blocking is compatible with $\rho < T$ for $\omega \leq 1 - T/2 = 0.0804$. For three rings the threshold is closed,

$$\rho_3^*(\omega) = \max\left(1, \min(2(1 - \omega), \max(\varphi, \frac{2}{1+2\omega}))\right),$$

with a purely additive proof and a *golden plateau* $\rho_3^* \equiv \varphi$ on $\omega \in [\frac{\sqrt{5}-2}{2}, 1 - \frac{\varphi}{2}]$; and a general additive tree argument shows $\rho_k^* = \rho_3^*$ for *every* $k \geq 3$. Consequently the combinatorial threshold crosses T exactly at

$$\omega_T = \frac{1}{T} - \frac{1}{2} = T^2 - T - \frac{3}{2} = 0.043689\dots$$

below ω_T the exchange step never blocks with $\rho < T$, with no geometry at all.

The canonical template with positive width. For $\omega > \omega_T$ geometry enters through the canonical template (v a diametral pan with one big neighbour α ; u the hole of α , capacity $\alpha - \omega$; $S = \{\sigma_1 \geq \sigma_2\}$ placed there by the witness). Blocking must defeat four resources, giving the walls $\sigma_2 > 1 - \omega$ (H_m), $\sigma_2 > \alpha - \omega - 1$ (σ_2 next to m inside u), $\sigma_1 + \sigma_2 \leq \alpha - \omega$ (the witness), and non-packability of the triple $\{\alpha, \sigma_1, \sigma_2\}$ in the pan. Two structural facts organize the optimum. First, on the blocking frontier of the triple one has the closed form $t(\sigma_1) + t(\sigma_2) = t(b(\alpha))$ with $t(s) = \sqrt{(1-s)/s}$ (the Descartes pocket read in the t -coordinate), and the frontier slope satisfies the α -*independent* identity

$$\kappa = -\frac{\partial \sigma_2}{\partial \sigma_1} = \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \quad g(s) = s^3(1-s), \quad \kappa \geq 1 \text{ for all } \alpha > 1,$$

so the cheapest blocked pair sits at $\sigma_1 \rightarrow 1, \sigma_2 \rightarrow b(\alpha)$; moreover $1 + b(\alpha) \geq \alpha$ iff $\alpha \leq T$, so no blocking exists at all for $\alpha \leq T$. Second, the witness capacity deforms the cubic: with T_c the positive root of $\alpha^3 = c(\alpha^2 + \alpha + 1)$, the witness branch has infimum $\Phi(\omega) = T_{1+\omega} - \omega$, strictly increasing and concave with $\Phi'(\omega) = (2\alpha + 1)/(\alpha^2(\alpha^2 + 2\alpha + 3))$ at $\alpha = T_{1+\omega}$. The full curve of the blocked infimum $T_{\text{can}}(\omega)$ is closed *as a proved lower bound* — the matching upper bound is exact on the witness branch and at the corner below, and inherits the usual angular-criterion caveat on the other two branches: $2(1 - \omega)$ on $(0, \omega_1]$, a sextic-algebraic middle branch on $[\omega_1, 1/7]$, and Φ on $[1/7, 0.30]$, where $\omega_1 = 0.0413570\dots$ is the root of $4\omega^3 - 20\omega^2 + 25\omega - 1$ in $(1/25, 1/14)$.

Theorem 20 (The 13/7 corner). *For all $\omega > 0$,*

$$T_{\text{can}}(\omega) \geq \frac{13}{7} = T + 0.017856\dots,$$

with equality exactly in the limit at the rational corner $(\omega, \alpha, \sigma_2) = (\frac{1}{7}, 2, \frac{6}{7})$, where all four walls saturate simultaneously (note $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7} \cdot 7$). The bound is unconditional — in particular it does not depend on the exactness of any angular feasibility criterion — and the approximating family is genuine. Thus positive width only raises the Tribonacci floor, uniformly.

The fine structure is amusing: the curve is *not* monotone on $(0, 1/7]$ — ω_1 is a local minimum, followed by a bump of height $1.1 \cdot 10^{-4}$ at $\omega_{\text{peak}} \approx 0.0445$ (a root of an explicit degree-8 polynomial) before descending to the corner.

8 Generic containers: coronas, walls, and metallic floors

The canonical template assumes v contains only the big neighbour and m . The remaining gap towards Conjecture 19 is genericity of the containers, and it is attacked by a toolkit of independent walls, each adversarially verified. Throughout, v is a pan of radius R with occupants $\{\alpha\} \cup O \cup \{m\}$, $O = \{o_1 \geq \dots \geq o_j\}$, $o_i \geq m = 1$, u is the hole of α , $S = \{\sigma_1 \geq \sigma_2\}$, and holes may be occupied arbitrarily, at any nesting depth.

The universal trio frontier. For a trio $\{A, x, y\}$ wall-tangent in a disk of *arbitrary* radius R (interior domain, and under the necessary hypothesis $A \geq \min(x, y)$), the angular blocking condition is *linear* in the coordinates $T_c(x) = \sqrt{(c-x)/x}$, $c = R - A$: θ -infeasibility $\iff T_c(x) + T_c(y) \leq \tau_R = c/\sqrt{AR}$. The pocket generalizes to $b_R(A) = ARc/(AR + c^2)$, increasing in R , and the frontier slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$, $g_c(s) = s^3(c-s)$, is uniform in R . Without the hypothesis $A \geq \min(x, y)$ the equivalence is false (an explicit counterexample pins the failure region exactly), a fact found by adversarial verification.

An exact corona criterion. Call a *corona* a packing of k circles all wall-tangent. For a fixed cyclic order the corona constraints form a linear feasibility system in the angular gaps, so coronas are decidable exactly; every subset of a corona yields a necessary *certificate* (its induced cyclic sum of pairwise angles is at most 2π). For $k = 4$ the certificates are exactly sufficient – for arbitrary symmetric separations in $(0, \pi]$, by a negative-cycle argument: a violating cycle with U ascents needs more than $2U$ edges, so on four nodes only $U = 1$ cycles can violate, and those are exactly the subset certificates – and the global criterion collapses to *two* inequalities: the top trio, plus the total of the *zigzag* order (a_1, a_3, a_2, a_4) , which minimizes the cyclic total by a convexity/majorization argument ($\theta = g(\log f(a) + \log f(b))$ with g convex). Two natural guesses are refuted by explicit counterexamples: the consecutive-arc condition alone is *not* sufficient for $k \geq 4$, and the sorted order is *not* optimal (the zigzag is: large circles must not be angular neighbours). For $k = 5$ sufficiency requires exactly one additional certificate, the *pentagram* (the five-diagonal star $\{5/2\}$, with combinatorially sharp example $\theta = \pi$ on the diagonals), conjecturally redundant for separations of geometric origin.

Occupant walls and metallic floors. Each extra occupant's hole is itself a reinsertion resource, and blocking it is *taxed*. The basic count: with free holes, blocking forces every extra occupant to within one width of m ($o_i < \sigma_2 + \omega$), and the largest occupant's tail gives $\rho > (j+2)/(1+\omega)$ – each occupant pays $1/(1+\omega)$ – whence, with a refined branch $\rho > 4/(1+2\omega)$ for $\omega \geq \frac{1}{2}$, $\rho > T$ for $\omega < 2/T - \frac{1}{2} = 0.5874$ when $j = 1$ and for *all* ω when $j \geq 2$; adding occupants is strictly suboptimal for the blocker, since the canonical curve satisfies $\Phi(\omega) < 2$ identically (the polynomial identity $(2+\omega)^3 - (1+\omega)((2+\omega)^2 + (2+\omega) + 1) = 1$). For occupied holes the tax is quantified by the *opposite-disk lemma*: in a disk of capacity c with largest piece σ , if the remaining mass is at most $c - \sigma$ the whole family packs (the wall-tangent σ leaves a free disk of radius exactly $c - \sigma$ opposite, filled in a row); hence blocking a hole costs at least its slack, $y < \sigma_2 + \omega + X_y$ with X_y the hole content, for every *extra* occupant o_i and every ring of radius $\geq r_m$ nested inside one (α and m themselves are excluded: α 's hole is u , governed by its own walls, and excluding m is essential). Following the *minimal* such ring and balancing two tails yields, for arbitrary hole occupancy at any depth,

$$\rho > \Psi(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + 1},$$

the metallic mean of $2(1-\omega)$: $\Psi(0) = 1 + \sqrt{2}$ is the silver ratio, $\Psi(1/4) = 2$ exactly, and $\Psi > T$ iff $\omega < (T-1)^2/2 = 0.3522$. Allowing m itself to carry children splits by an evacuation dichotomy into the same bound and a second metallic mean, the root of $u^2 - (2-\omega)u - 1$, which dominates automatically; its own crossing is at $(T-1)^2$ exactly – twice the previous one, and the certifying identity is the Tribonacci polynomial: $(T-1)^2 T - (2T - T^2 + 1) = T^3 - T^2 - T - 1$.

The double-pocket wall and the golden line. The geometric wall needs no angular criterion at all: blocking implies the four rings $\{\alpha, o_1, \sigma_1, \sigma_2\}$ do not pack in the pan, hence not in the disk of radius $\bar{R} = \alpha + o_1$ (containment), where the pair $\{\alpha, o_1\}$ is diametrically rigid and leaves exactly *two* Descartes pockets $b_2(\alpha, o_1) = \alpha o_1(\alpha + o_1)/(\alpha^2 + \alpha o_1 + o_1^2)$, one on each side (their mirror centers are $4b_2$ apart: the exact identity $y_0 = 2b_2$). Thus blocking forces $\sigma_1 > b_2(\alpha, o_1)$. Feeding this wall into the program of the previous walls produces a *golden* optimum: the binding corner is $\alpha = 2$, $o_1 = \sqrt{5} - 1$ – the pair with unit pocket, $b_2(2, \sqrt{5} - 1) = 1$ exactly, self-dual in the sense $A_{\max}(\sqrt{5} - 1) = 2$, $A_{\max}(2) = \sqrt{5} - 1$ – and, for one extra occupant and *any* hole occupancy, both branches of the analysis meet the same line: $\rho > T$ unconditionally for all $\omega < \omega_A$, via

$$\rho > \varphi^2 - \frac{\varphi}{2}\omega, \quad \omega_A = 2 - 2(T-1)(\varphi - 1) = 0.962585\dots$$

(the golden line itself is proved in all but one corner configuration – branch B with a nested node inside the occupant and $\alpha < o_1$, possible only for $\omega < \frac{1}{2}$ – where the proved bound is $\geq 2 > T$, via $\Psi_B(\frac{1}{2}) = 2$ exactly). The proof combines two exact facts – b_2 is concave in α ($\partial^2 b_2 / \partial \alpha^2 = -6\alpha o^3(\alpha + o)/D^3$) and $D^2 - \alpha o^2(o + 2\alpha) = (\alpha + o)(\alpha^3 + \alpha^2 o + o^3) > 0$ – with one-variable boundary certificates anchored at the identity $f_1 \equiv$ curve at $o_1 = \sqrt{5} - 1$. For $j \geq 2$ occupants the ladder $\Psi_j = (1 - \omega) + \sqrt{(1 - \omega)^2 + j}$ (crossing T at $1 - (T^2 - j)/(2T)$: 0.624 for $j = 2$, 0.896 for $j = 3$, all ω for $j \geq 4$) covers the corresponding ranges, modulo a pending short recursion patch for the case where the largest occupant’s hole itself contains a ring of radius $\geq r_m$ (numerically robust with wide margins; $\Psi_1 = \Psi$ is unconditional). Small rings anywhere compatible with each theorem’s template – and with S still a pair – are free for all the combinatorial walls, whose unblocking placements are local to D_m , H_m , the holes, and the slot next to m inside u .

Status. Together with Theorem 20, every blocked exchange in the templates above satisfies $\rho > T$: for all ω in the canonical template, for $\omega < 0.9626$ with one extra occupant under arbitrary hole occupancy, for $\omega < 0.624$ and $\omega < 0.896$ with two and three (modulo the Ψ_j patch above), and for all ω with four or more. What remains open on the road to Conjecture 19 is documented in the repository: the width tips ($j = 1$, $\omega \geq 0.9626$; $j = 2$, $\omega \geq 0.624$; $j = 3$, $\omega \geq 0.896$), the Ψ_j recursion patch, a quantitative “gap lemma” (small rings against the geometric wall; tiny σ_2), profiles S with three or more pieces beyond the combinatorial bound, and the second battle where u is the pan itself. Each statement in this section is backed by a verification script and an adversarial verification report in the repository.

9 The two objectives diverge: a phase diagram

Contact area behaves like $2\pi w \sum r_i$ minus a per-ring penalty πw^2 ; cardinality counts rings. With uniform width the hole of a large ring is large, so nesting usually reconciles the two objectives, and the divergence survives only marginally: small rings that fit k times in the pan but $k - 1$

times in the large ring's hole. The minimal instance is $R = 10$, $w = 1$, radii $\{9.0, 4.2, 4.2, 4.2\}$: the area optimum is the 9.0 with one nested 4.2 ($N = 2$, $A \approx 76.7$), the cardinality optimum the three 4.2's ($N = 3$, $A \approx 69.7$). Figure 1 maps the divergence region for the family “one large ring b plus unlimited equal small rings ρ ”: a staircase governed by the proved optimal thresholds for n equal circles in a disk [8, 9, 10, 11, 12]; its upper edge is exactly the three-circle threshold $0.4641R$.

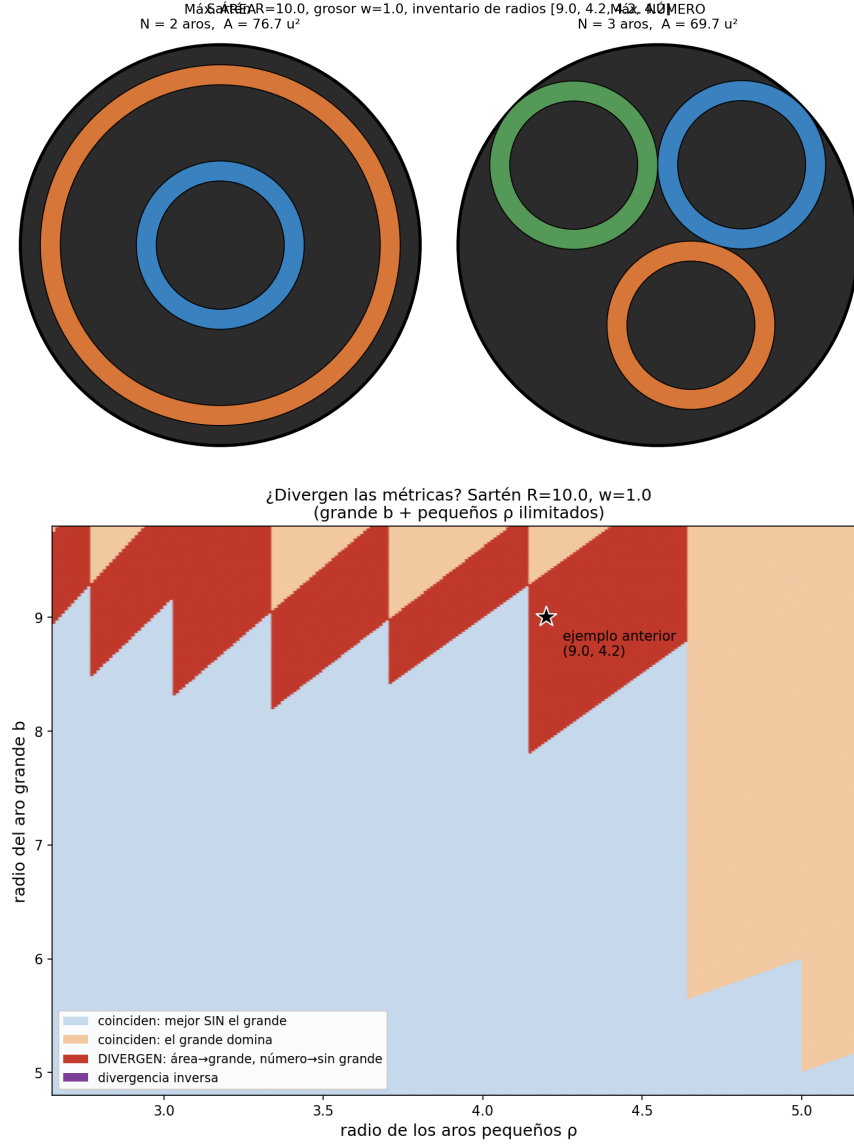


Figure 1: Top: the minimal divergence instance; area optimum (left) versus cardinality optimum (right). Bottom: phase diagram of the divergence band for uniform width.

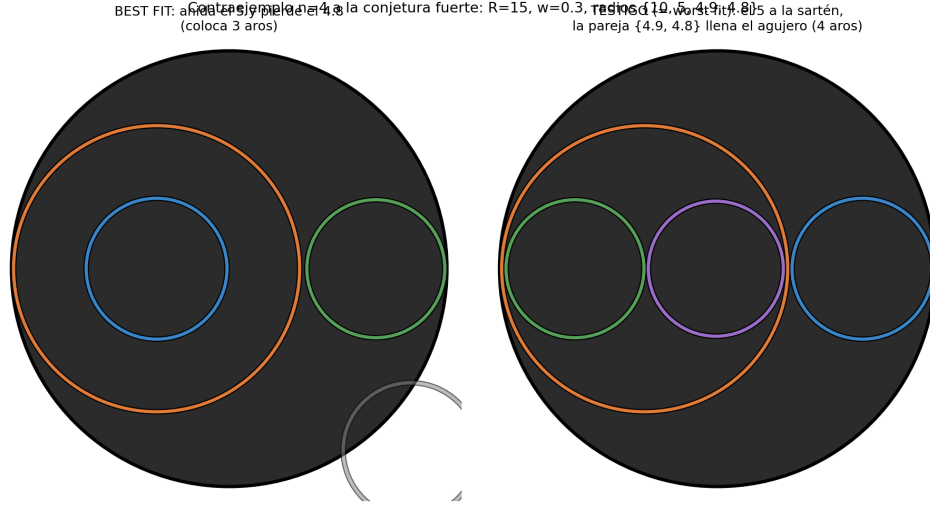


Figure 2: The $n = 4$ counterexample of Theorem 13: best fit (left) nests the 5 and loses the 4.8; the witness (right), realized by worst fit, places all four rings.

10 Hardness decouples into two layers

If all radii lie within a band of width less than w , no nesting is possible and the decision problem “do all rings fit?” is exactly circle packing of the outer radii into the pan.

The geometric layer. Since Theorems 6 and 9 hold for arbitrary containers, our model contains, as the special case of a *square* pan with a no-nesting band, the problem of deciding whether given circles fit into a square, which is NP-hard by reduction from 3-PARTITION [4]; hence the geometric layer of the model is NP-hard. Membership in NP is itself delicate, because tight packings may require coordinates without short descriptions – a phenomenon made precise by the $\exists\mathbb{R}$ -completeness framework for two-dimensional packing [6]. For the disk pan specifically, we are not aware of a published NP-hardness proof (the related-work discussions of [5] and its companions attribute hardness to the square container only), and we leave its precise status as part of Open Problem 24. Integer-programming formulations for circle packing in rectangular containers are studied in [16].

The combinatorial layer. Independently of the geometry, the *additive surrogate* (Theorem 16) is weakly NP-hard, and here the reduction can be made exact despite the per-ring penalty in the area.

Proposition 21 (The additive layer is weakly NP-hard). *Deciding whether an instance of the additive surrogate admits a feasible set of contact area at least a given threshold is weakly NP-hard, already for instances with a no-nesting radius band.*

Proof. Reduce from EQUAL-CARDINALITY PARTITION: given positive integers $a_1, \dots, a_n$, n even, with $\sum_i a_i = 2B$, decide whether some S with $|S| = n/2$ has $\sum_S a_i = B$. (This variant is NP-hard by the standard padding: scaling a PARTITION instance by $2n$ and adjoining n unit items forces, modulo $2n$, every solution to use exactly n of the $2n$ items.) Fix any width $w > 0$, set $M = 3B$, pick $\delta \in (\frac{w}{3B}, \frac{w}{2B})$, and take radii $r_i = \delta(a_i + M)$ with pan capacity

$R^+ = \delta(\frac{n}{2}M + B)$. All radii lie in $[3B\delta, 5B\delta]$, so $r_i > w$ (each is a genuine ring, $a(r) = 2\pi wr - \pi w^2$) and $r_{\max} - r_{\min} = 2B\delta < w$ (no nesting): feasibility is exactly $\sum_S r_i \leq R^+$. Write $s = |S|$ and $t = \sum_S a_i \leq 2B$; the area is $A(S) = 2\pi w\delta(sM + t) - \pi w^2 s$, and we ask whether $A(S) \geq \Theta := 2\pi w\delta(\frac{n}{2}M + B) - \pi w^2 \frac{n}{2}$. If a balanced partition exists, taking S with $s = n/2$, $t = B$ attains Θ exactly. Conversely: $s > n/2$ is infeasible, since it would force $(s - \frac{n}{2})M \leq B - t \leq B < M$. For $s = \frac{n}{2} - q$ with $q \geq 1$, the requirement $A(S) \geq \Theta$ rearranges to $wq \geq 2\delta(qM + B - t)$, but

$$2\delta(qM + B - t) \geq 2\delta(qM - B) \geq 2\delta q(M - B) = 4\delta qB > wq$$

using $t \leq 2B$, $q \geq 1$, $M = 3B$ and $\delta > \frac{w}{4B}$ (implied by $\delta > \frac{w}{3B}$): impossible. Finally $s = n/2$ forces $t \geq B$ from $A \geq \Theta$ and $t \leq B$ from feasibility, i.e. $t = B$: a balanced partition. $\square$

Superincreasing radii eliminate exactly this combinatorial layer – precisely as in the superincreasing knapsack [15] – leaving only the geometric one: by Theorem 6, the area optimum then reduces to n calls to a sibling-packing feasibility oracle. Useful blackboxes for the remaining geometry include the critical-density theorem for disks in a disk (any family of total area at most half the container’s packs, and $1/2$ is tight) [5] and Split Packing [7].

11 Generalizations

The model invites several generalizations, all of which we record here both as context for the theorems (two of them hold with no extra work) and as a research program.

(a) *Container shape and dimension.* Theorems 6 and 9 are stated and proved for arbitrary containers in $\mathbb{R}^d$: rectangular griddles, and tubes/spherical shells in three dimensions – the original RCPP setting – are covered verbatim, because the exchange argument acts inside the vacated ball of the moved ring only. The sharpness results (Section 5) are disk-specific; their analogues for squares and in $\mathbb{R}^3$ are open.

(b) *Variable width.* With per-ring widths w_i , a large ring may have a small hole, which destroys the “nesting rescue” and should widen the divergence band of Section 9 dramatically; the superadditivity lemma must be re-examined since a depends on both r and w .

(c) *Flexible rings.* Real squid rings bend: a ring partially resting on another still touches the pan outside the overlap, up to a lifted ramp of width δ around each overlap. The idealization $\delta = 0$ (contact area equals the annulus minus the overlapped region) defines a continuous relaxation in which partial placements trade contact for cardinality; $\delta > 0$ penalizes overlaps by a dead band proportional to the overlap perimeter.

(d) *Infinite inventories.* Unlimited supplies (already used in the phase diagram) suggest degenerate limits: maximal nested chains $r, r - w, r - 2w, \dots$, density questions, and Apollonian-type limit configurations.

(e) *Objectives.* Theorem 6 covers every positive, strictly increasing, superadditive value; weighted counts and concave values fall outside and, by Proposition 8, genuinely so.

12 Open problems

Open Problem 22. Prove Conjecture 19 by assembling the universal reinsertion lemma from the pieces of Sections 7–8: combinatorics ($\rho_k^* = \rho_3^*$, $\omega_c = \omega_T$) below ω_T and the wall hierarchy above it. The documented remaining gaps are the width tips ($j = 1$ with $\omega \geq 0.9626$; $j = 2$ with

$\omega \geq 0.624$; $j = 3$ with $\omega \geq 0.896$), the Ψ_j recursion patch, a quantitative gap lemma (small rings against the double-pocket wall; tiny σ_2), profiles with three or more reinserted pieces beyond the combinatorial bound, and the case where u is the pan.

Open Problem 23. Can a placement rule with access to the full input attain the lex-max set with polynomially many calls to a sibling-packing oracle, or is there a complexity-theoretic obstruction?

Open Problem 24. Characterize the divergence band of Section 9 exactly; determine the approximability of the cardinality objective; decide whether the additive surrogate admits a pseudopolynomial algorithm for integer radii; and settle the complexity of deciding packability of given circles into a *disk* container (NP-hardness is only published for the square).

Acknowledgements

The results of Sections 6–8 were developed with extensive computer assistance (Anthropic’s Claude) under a draft–verify workflow: every result was independently rederived from its bare statement by an adversarial verification pass before being accepted, and several statements were corrected or sharpened by that pass; the reports are part of the repository. To be explicit about epistemic status: the mathematical guarantee for every claim is the written proof and, where a proof delegates an identity, an exact symbolic computation; the verification workflow and the numerical sweeps are quality control and evidence, never a substitute for proof.

A Verification map

Every theorem of Sections 6–8 is backed in the accompanying repository by (i) a self-contained draft with complete proofs, (ii) a verification script whose blocks separate exact symbolic identities (**sympy**) from numerical sweeps, all green, and (iii) an *adversarial verification report*: an independent rederivation of each result from its statement alone, followed by an audit and directed attacks. The correspondence, as “verified claims (script blocks)”, is: Theorem 17 – **suelo_rigido**, 10/10 claims (7 blocks); slack pans and the universal trio frontier – **universal** (5/5 blocks); the κ identity – **h1**, 6/6 claims (5/5 blocks); the width curve and Theorem 20 – **grosor** 8/8 claims, **esquina** 6/6 claims (5/5 blocks); profile thresholds – **perfil_tres**, **cuatro** (**cuatrok** 5/5 blocks); the corona criterion – **corona** (5/5 blocks); occupant walls, the opposite-disk lemma and the metallic floors – **ocupantes** (5/5), **bloqueadores** (6/6); the double-pocket wall and the golden line – **bolsillo** (6/6). Several statements were corrected or sharpened by the adversarial round (a missing hypothesis in the trio frontier, a refuted “leak” family, the $\alpha \geq o_1$ case split); the reports record each incident.

B Complete proof of Theorem 17

Throughout, work in the normalization of Theorem 17: $r_1 = 1$, $t = r_2$, $u = r_3$, $v = r_4$, width $\omega > 0$, pan radius $R = 1 + t$, and recall $b(t) = t(1 + t)/(1 + t + t^2)$, $T^3 = T^2 + T + 1$, $t^* = 1/T$ (equivalently $t^{*3} + t^{*2} + t^* = 1$), $\varphi = (1 + \sqrt{5})/2$. All exact algebraic identities used below are additionally verified in **code/rigido.py** (block V5).

Lemma 25 (Half-angle identity). *Let two circles of radii a, b be internally tangent to the wall of a disk of radius R (centers at distances $R - a, R - b$ from the center). If $a + b \leq R$, the minimal angular separation $\theta(a, b)$ guaranteeing disjoint interiors is well defined, lies in $[0, \pi]$, satisfies*

$$\sin^2 \frac{\theta(a, b)}{2} = f(a)f(b), \quad f(x) = \frac{x}{R - x},$$

and is increasing in each argument; two wall-tangent circles at angular separation γ have disjoint interiors iff $\gamma \geq \theta(a, b)$.

Proof. By the law of cosines the center distance $D(\gamma)^2 = (R - a)^2 + (R - b)^2 - 2(R - a)(R - b) \cos \gamma$ is increasing on $[0, \pi]$, and disjointness is $D \geq a + b$. At the threshold, $1 - \cos \theta = [(a + b)^2 - (a - b)^2] / (2(R - a)(R - b)) = 2ab / ((R - a)(R - b))$, i.e. $\sin^2(\theta/2) = f(a)f(b)$; and $f(a)f(b) \leq 1$ is exactly $a + b \leq R$. Monotonicity is that of f . $\square$

Lemma 26 (Construction). *Let $v \leq u \leq t < 1$ and $R = 1 + t$. If $\theta(1, u) + \theta(1, v) + \theta(u, v) \leq 2\pi$, then $\{1, u, v\}$ packs in the disk of radius R .*

Proof. All three angles are defined ($1 + u \leq R \iff u \leq t$, and $u + v \leq 2t < R$). Place the three circles wall-tangent with the unit circle in the middle: centers at angles $0, +\theta(1, u), -\theta(1, v)$. Each circle is contained in the disk (internal wall tangency). The pairs $(1, u)$ and $(1, v)$ are separated by exactly their thresholds. For (u, v) , the angular difference is $\Delta = \theta(1, u) + \theta(1, v)$ and the separation is $\gamma = \min(\Delta, 2\pi - \Delta)$. If $\gamma = \Delta$: by monotonicity (Lemma 25, $1 \geq u$), $\theta(1, v) \geq \theta(u, v)$, so $\Delta \geq \theta(u, v)$. If $\gamma = 2\pi - \Delta$: the hypothesis gives $2\pi - \Delta \geq \theta(u, v)$. $\square$

Lemma 27 (Algebraic reduction). *Define $\psi(x) = \sqrt{(t - x)/x}$ (decreasing, $\psi(t) = 0$) and $\tau = t/\sqrt{1 + t}$; note the exact identities $\psi(b(t)) = \tau$ and $t/(1 + \tau^2) = b(t)$. If $v \leq u \leq t < 1$ and $\psi(u) + \psi(v) \geq \tau$, then $\theta(1, u) + \theta(1, v) + \theta(u, v) \leq 2\pi$, and hence $\{1, u, v\}$ packs in the disk $R = 1 + t$ by Lemma 26.*

Proof. Let $A = \sqrt{f(1)f(u)}$, $B = \sqrt{f(1)f(v)}$, $C = \sqrt{f(u)f(v)}$ (the sines of the half-angles); the goal is $\arcsin A + \arcsin B + \arcsin C \leq \pi$. If $\arcsin A + \arcsin B \leq \pi/2$ there is nothing to prove, since $\arcsin C \leq \pi/2$. Otherwise the goal is equivalent, both sides lying in $[0, \pi/2]$, to $C \leq \sin(\arcsin A + \arcsin B) = A\sqrt{1 - B^2} + B\sqrt{1 - A^2}$. Dividing by $C > 0$ and writing $G(x) = f(1)/f(x) - f(1)^2$,

$$\frac{A\sqrt{1 - B^2}}{C} = \sqrt{G(v)}, \quad \frac{B\sqrt{1 - A^2}}{C} = \sqrt{G(u)}, \quad G(x) = \frac{(1 + t)(t - x)}{t^2 x},$$

the last identity by direct computation with $f(1) = 1/t$, $f(x) = x/(1 + t - x)$; the radicals are real because $f(1)f(x) \leq 1 \iff x \leq t$. Thus $\sqrt{G(x)} = (\sqrt{1 + t}/t)\psi(x)$ and, by hypothesis, $\sqrt{G(u)} + \sqrt{G(v)} = (\sqrt{1 + t}/t)(\psi(u) + \psi(v)) \geq (\sqrt{1 + t}/t)\tau = 1$, which is the required inequality. $\square$

The contrapositive used below: hypothesis (F3) implies $\psi(u) + \psi(v) < \tau$. This is the only direction ever used – the *constructive* one – so no exactness of any angular criterion is assumed anywhere in this appendix.

Lemma 28 (Corner minimization). *Let $0 < v \leq u \leq t < 1$ with $\psi(u) + \psi(v) < \tau$. Then: (1) $u, v > b(t)$; (2) if $t \geq 1/\varphi$ then $u + v > 1$; (3) if $t < 1/\varphi$ then $u + v > t + b(t)$.*

Proof. (1) Each summand is $< \tau$, ψ is decreasing and $\psi(b(t)) = \tau$. (2) $u + v > 2b(t)$ and $2b(t) - 1 = (t^2 + t - 1)/(t^2 + t + 1) \geq 0$ exactly when $t \geq 1/\varphi$. (3) Substitute $\alpha = \psi(u)$, $\beta = \psi(v)$, so $u = U(\alpha)$, $v = U(\beta)$ with $U(z) = t/(1 + z^2)$ strictly decreasing; the region is $\alpha, \beta \geq 0$, $\alpha + \beta < \tau$. Increasing α to $\tau - \beta$ decreases u , so $u + v > U(\tau - \beta) + U(\beta) =: W(\tau - \beta)$. Now $U''(z) = 2t(3z^2 - 1)/(1 + z^2)^3 \leq 0$ for $z \leq 1/\sqrt{3}$, and $\tau \leq 1/\sqrt{3} \iff 3t^2 - t - 1 \leq 0 \iff t \leq (1 + \sqrt{13})/6 = 0.7676\dots$, which covers $t < 1/\varphi = 0.618\dots$ with room to spare. Hence U is concave on $[0, \tau]$, W is concave, and its minimum over $[0, \tau]$ is at the endpoints: $W \geq W(0) = U(0) + U(\tau) = t + t/(1 + \tau^2) = t + b(t)$, by the identity $t/(1 + \tau^2) = b(t)$. $\square$

Proof of the strict floor in Theorem 17. Let $I \in \mathcal{F}$, normalized. By (F3) and the contrapositive of Lemmas 26–27, $\psi(u) + \psi(v) < \tau$. If $t \geq 1/\varphi$, Lemma 28(2) gives $u + v > 1$, contradicting (F2) ($u + v \leq 1 - \omega < 1$); so $t < 1/\varphi$ and Lemma 28(3) gives the blocking pressure $u + v > t + b(t)$. Chaining with the witness pressure (F2),

$$t + b(t) < u + v \leq 1 - \omega < 1,$$

and $t + b(t) < 1 \iff t(1 + t) < (1 - t)(1 + t + t^2) = 1 - t^3 \iff t^3 + t^2 + t < 1 \iff t < t^*$ (which also proves that $\mathcal{F}$ forces $t < t^*$). Finally

$$\rho \geq \rho_2 = \frac{u + v}{t} > \frac{t + b(t)}{t} = 1 + \frac{1 + t}{1 + t + t^2} =: L(t),$$

and L is strictly decreasing ($L' = -(t^2 + 2t)/(1 + t + t^2)^2 < 0$) with $L(t^*) = 1/t^* = T$ (an identity modulo $t^{*3} + t^{*2} + t^* = 1$). Since $t < t^*$: $\rho > L(t) > L(t^*) = T$. $\square$

It remains to prove exactness of the infimum. Two ingredients: the rigid pocket is exact, and infeasibility survives small perturbations.

Proposition 29 (Exact rigid pocket). *For $0 < v \leq t < 1$, the triple $\{1, t, v\}$ packs in the disk of radius $R = 1 + t$ if and only if $v \leq b(t)$.*

Proof. Rigidity. Centers satisfy $|c_1| \leq t$, $|c_t| \leq 1$ and $|c_1 - c_t| \geq 1 + t$; the triangle inequality forces equality throughout, so up to rotation $c_1 = (-t, 0)$, $c_t = (1, 0)$. *Sufficiency.* The pocket circle tangent to the wall and to both has radius $b(t)$; any $v \leq b(t)$ placed concentrically inside it works. *Necessity.* Let X be the center of v , $d = |X| > 0$, and γ the angle between X and $(1, 0)$. Disjointness reads $d^2 + t^2 + 2dt \cos \gamma \geq (1 + v)^2$ and $d^2 + 1 - 2d \cos \gamma \geq (t + v)^2$, which bound $\cos \gamma$ from below and above; compatibility of the two bounds requires $(1 + t)d^2 \geq (1 + v)^2 + t(t + v)^2 - t(1 + t)$, and with $d \leq R - v = 1 + t - v$:

$$(1 + t)(1 + t - v)^2 - (1 + v)^2 - t(t + v)^2 + t(1 + t) \geq 0.$$

The left-hand side factors *exactly* as $4(t^2 + t + 1)(b(t) - v)$ (symbolic identity, V5), whence $v \leq b(t)$. $\square$

Lemma 30 (Closure and monotonicity of packability). *Let $D = \{(t, u, v) : 0 < v \leq u \leq t < 1\}$ and say $p = (t, u, v)$ packs if there are centers with $|c_1| \leq t$, $|c_2| \leq 1 + t - u$, $|c_3| \leq 1 + t - v$, $|c_1 - c_2| \geq 1 + u$, $|c_1 - c_3| \geq 1 + v$, $|c_2 - c_3| \geq u + v$. Let $E \subseteq D$ be the packable set. Then: (1) E is downward closed in (u, v) ; (2) E is closed in D ; (3) for fixed t, v , the set $\{u \in [v, t] : (t, u, v) \in E\}$ is empty or a closed interval $[v, u_{\max}]$ with the maximum attained; in particular, if $(t, t, v) \notin E$, then $(t, u, v) \notin E$ for all $u \in (u_{\max}, t]$ – the infeasibility of the rigid corner propagates to $\delta := t - u \in [0, \delta_0)$, $\delta_0 = t - u_{\max} > 0$.*

Proof. (1) The same centers witness (all six constraints relax). (2) Witness centers live in a fixed compact ball; take a convergent subsequence of witnesses along $p_k \rightarrow p$ and pass to the limit in the six non-strict continuous constraints. (3) By (1) the set is an initial segment, by (2) it is closed. $\square$

Proposition 31 (The infimum is T and is not attained). $\inf\{\rho(I) : I \in \mathcal{F}\} = T$.

Proof. $\geq$ and non-attainment: the strict floor above. $\leq$: fix $t < t^*$, set $\varepsilon_t = (1 - t - b(t))/2 > 0$ and $v_t = b(t) + \min(\varepsilon_t, (t - b(t))/2)$. By Proposition 29, (t, t, v_t) is infeasible; by Lemma 30(3) there is $\delta \in (0, \min(\varepsilon_t, t - v_t))$ with $(t, t - \delta, v_t)$ infeasible. The instance $I_t = (1, t, t - \delta, v_t)$ with $\omega = 1 - (t - \delta + v_t) > 0$ satisfies (F1)–(F3) and the strict orderings, so $I_t \in \mathcal{F}$, and $\rho(I_t) = \rho_2 \leq (t + b(t) + \varepsilon_t)/t = L(t) + \varepsilon_t/t \rightarrow L(t^*) = T$ as $t \rightarrow t^*$ (the other tails are smaller: $\rho_1 = t + u + v < 3t < T$ and $\rho_3 = v/u < 1$). $\square$

Proof of Remark 18. A packing in a disk of radius $R' \leq R$ is a packing in the disk of radius R (translate to concentric position; containment is preserved, pairwise constraints are unchanged). Hence non-packability at radius R implies non-packability at radius $r_1 + r_2 \leq R$, so an instance with a slack pan satisfying (F2) and (F3) at its own radius satisfies (F2)–(F3) at radius $r_1 + r_2$, and the floor applies verbatim; the approximating family of Proposition 31 has $R = r_1 + r_2$, which is admissible, so the infimum is unchanged. $\square$

C Verification details for Theorem 14

All feasibilities in the twin instances are two-circle-exact except three facts, proved here; every numerical claim is also reproduced by the scripts in the repository (`code/gemelas.py`).

(V1) $\{10, 4.99, 4.50\}$ does not pack in a disk of radius 15. As in Lemma 12: $|c_A| \leq 5$, $|c_X| \leq 10.01$, $|c_Y| \leq 10.5$, $|c_A - c_X| \geq 14.99$, $|c_A - c_Y| \geq 14.5$, $|c_X - c_Y| \geq 9.49$; the first two give $a := |c_A| \geq 4.98$. With $u = -c_A/|c_A|$, $c_X \cdot u \geq (124.5 - a^2)/(2a) \geq 9.95$ and $c_Y \cdot u \geq (100 - a^2)/(2a) \geq 7.5$ (both decreasing in a , evaluated at $a = 5$), with transverse norms at most $\sqrt{10.01^2 - 9.95^2} \leq 1.095$ and $\sqrt{10.5^2 - 7.5^2} \leq 7.349$. Hence $|c_X - c_Y|^2 \leq 10.01^2 + 10.5^2 - 2(9.95 \cdot 7.5 - 1.095 \cdot 7.349) \leq 77.30$, so $|c_X - c_Y| \leq 8.80 < 9.49$.

(V2) With the pan pair $\{10, 5\}$ placed, no third circle of radius $z \in \{4.74, 4.76\}$ fits. From $|c_A| \leq 5$, $|c_B| \leq 10$ and $|c_A - c_B| \geq 15$ the triangle inequality forces equalities: the pair is diametrically rigid, $c_B = 10u$ for a unit vector u and $c_A = -5u$. For a third circle of radius z : $c_z \cdot u \geq ((10+z)^2 - (15-z)^2 - 25)/10$, which equals 8.8 for $z = 4.76$ and 8.7 for $z = 4.74$; then $|c_z - c_B|^2 = |c_z|^2 - 20 c_z \cdot u + 100 \leq (15 - z)^2 - 20 c_z \cdot u + 100$, giving $|c_z - c_B| \leq 5.38 < 9.76$ and $\leq 5.60 < 9.74$ respectively.

(V3) $\{10, 4.76, 4.74\}$ packs in a disk of radius 15. Explicit boundary configuration: $c_A = (-5, 0)$, $c_X = 10.24(\cos 30.77^\circ, \sin 30.77^\circ) \approx (8.799, 5.238)$, $c_Y = 10.26(\cos 32.02^\circ, -\sin 32.02^\circ) \approx (8.700, -5.440)$. Then $|c_A| = 5$, $|c_X| = 10.24 = 15 - 4.76$, $|c_Y| = 10.26 = 15 - 4.74$ (all inside); $|c_A - c_X| = 14.76$ and $|c_A - c_Y| = 14.74$ (exact tangencies, chosen via the boundary angles); and $|c_X - c_Y| \approx 10.68 \geq 9.50$ with ample margin.

(V4) Hole facts. The hole of the 10 has radius 9.495; $4.99 + 4.50 = 9.49 \leq 9.495$ (the pair fits, two-circle exact) while $4.76 + 4.74 = 9.50 > 9.495$ and $5 + 4.50 = 9.50 > 9.495$ (blocked, two-circle exact).

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